



Problem Statement

You are given a grid of dimensions $N \times M$. Each cell contains one of the following values:
 $0 \rightarrow$ Empty cell $1 \rightarrow$ Fresh orange $2 \rightarrow$ Rotten orange
A rotten orange can rot an adjacent fresh orange in 1 unit of time.
A rotten orange at position (i, j) can rot fresh oranges at:
 $(i-1, j) \rightarrow$ Up $(i+1, j) \rightarrow$ Down $(i, j-1) \rightarrow$ Left $(i, j+1) \rightarrow$ Right
Find the minimum time required to rot all fresh oranges.
If it is impossible to rot all fresh oranges, print -1.
If there are no fresh oranges initially, the required time is 0.

Input Format

The first line contains an integer T , the number of test cases.
For each test case: The first line contains two integers N and M , representing the number of rows and columns. The next N lines contain M space-separated integers representing the grid.

Constraints

$1 \leq T \leq 100$ $1 \leq N, M \leq 500$ $grid[i][j] \in \{0, 1, 2\}$

Output Format

Print a single integer representing the minimum time required to rot all fresh oranges.
Print -1 if some fresh oranges can never become rotten

Sample Input 0

3 3
2 1 1
1 1 0
0 1 1

Sample Output 0

4

Sample Input 1

3 3
2 1 1
0 1 1
1 0 1

Sample Output 1

-1

Change Theme

Language Python 3

```
1 from collections import deque
2 n, m = map(int, input().split())
3 grid = [list(map(int, input().split())) for _ in range(n)]
4 q = deque()
5 fresh = 0
6 for i in range(n):
7     for j in range(m):
8         if grid[i][j] == 2:
9             q.append((i, j))
10        elif grid[i][j] == 1:
11            fresh += 1
12
13 time = 0
14 while q and fresh:
15     for _ in range(len(q)):
16         i, j = q.popleft()
17         for di, dj in [(-1,0),(1,0),(0,-1),(0,1)]:
18             ni, nj = i + di, j + dj
19             if 0 <= ni < n and 0 <= nj < m and grid[ni][nj] == 1:
20                 grid[ni][nj] = 2
21                 q.append((ni, nj))
22                 fresh -= 1
23     time += 1
24 print(time if fresh == 0 else -1)
```

Line: 24 Col: 1

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